

Lab Toolkit Reference

Every tool used in the CompTIA Certified A+ Training (Core 1 and Core 2) labs. All are free, all run in the browser, and none requires an install or a login.

Browser tools

Tool	Link	What it does
IP Calculator	https://alfredang.github.io/ipcalculator/	Browser subnet calculator for IPv4 and IPv6 — CIDR, netmask, network and broadcast addresses, usable host ranges and batch processing with CSV export.
PCAP Analyzer	https://alfredang.github.io/pcapanalyzer/	Browser packet-capture analyser — protocol distribution, top talkers, top conversations and a packet table with per-packet detail. Files are parsed locally and never uploaded.
Cybersecurity Simulator	https://alfredang.github.io/cybersecuritysimulator/	Safe threat-simulation lab covering phishing, XSS, SQL injection, password strength, malware, ransomware, social engineering and data leakage.
RegexLab	https://alfredang.github.io/regexgenerator/	Live regular-expression tester with flags, match explanation and substitution — used to filter and parse support logs.
Killercoda Ubuntu Playground	https://killercoda.com/playgrounds/scenario/ubuntu	Free browser-based Ubuntu terminal with root access — no install required. Used for every Linux command-line lab in this course.

Which lab uses which tool

Tool	Labs
IP Calculator	Lab 3, Lab 4, Lab 6, Lab 9, Lab 28, Lab 42
PCAP Analyzer	Lab 7, Lab 11, Lab 28
Cybersecurity Simulator	Lab 38, Lab 39, Lab 40, Lab 41, Lab 47
RegexLab	Lab 46
Killercoda Ubuntu Playground	Lab 5, Lab 6, Lab 9, Lab 11, Lab 14, Lab 20, Lab 22, Lab 24, Lab 28, Lab 34, Lab 35, Lab 46, Lab 49, Lab 52, Lab 54

Killercoda Ubuntu Playground

A real Ubuntu machine with root access, running in a browser tab. Used for every Linux command-line lab in this course.

```
# Every Killercoda lab starts by refreshing the package index
apt-get update -qq
apt-get install -y <the packages that lab needs>
```

The playground resets when idle. If your terminal disappears, reopen it and re-run the setup commands from the start of the lab.

Windows command reference

```
ipconfig /all
ipconfig /release
ipconfig /renew
ipconfig /flushdns
ping <host>
tracert <host>
nslookup <domain>
netstat -ano
sfc /scannow
DISM /Online /Cleanup-Image /RestoreHealth
chkdsk C: /scan
robocopy <src> <dst> /MIR /R:2 /W:2
net accounts
net user
net localgroup Administrators
manage-bde -status C:
netsh advfirewall show allprofiles state
taskmgr
```

```
msconfig
```

```
eventvwr.msc
```

```
diskmgmt.msc
```

```
devmgmt.msc
```

```
taskschd.msc
```

```
lusrmgr.msc
```

```
perfmon.msc
```

```
gpedit.msc
```

```
cleanmgr
```

```
dfrgui
```

```
resmon
```

```
msinfo32
```

Linux command reference

```
pwd
```

```
ls -la
```

```
cd <dir>
```

```
cp <src> <dst>
```

```
mv <src> <dst>
```

```
rm -r <dir>
```

```
mkdir -p <dir>
```

```
cat <file>
```

```
less <file>
```

```
grep -n '<pattern>' <file>
```

```
find <path> -name '<glob>'
```

```
df -h
```

```
du -sh <dir>
```

free -h

ps aux

top

htop

chmod 750 <file>

chown user:group <file>

useradd -m <user>

usermod -aG sudo <user>

ip -brief addr show

ip route show

ping -c 3 <host>

dig +short <domain>

traceroute <host>

ss -tulnp

tcpdump -i any -c 100 -w capture.pcap

apt-get update && apt-get install -y <pkg>

smartctl -H /dev/sda

lsblk

man <command>